Lecture 4: Laplace Equation

In this lecture, we only consider the Laplace equation in three dimensional space, i.e., in (x, y, z)-space. For a bounded connected domain Ω with smooth boundary $\partial\Omega$ and smooth functions f and h, we consider the following problems

$$\boxed{\begin{cases} \triangle u = f, & \text{in } \Omega \\ u = h, & \text{on } \partial\Omega. \end{cases}} \tag{4.1}$$

If $f = 0$, then we call the solution u is a harmonic function. The arguments of two dimensional or higher dimensional case are similar.

4.1 Green's First Identity

First let us recall the notation (in three dimensions)

$$\text{grad} f = \nabla f = \text{the vector } (f_x, f_y, f_z)$$

$$\text{div} F = \nabla \cdot F = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z},$$

where $F = (F_1, F_2, F_3)$ is a vector field. Also,

$$\triangle u = \text{divgrad} u = \nabla \cdot \nabla u = u_{xx} + u_{yy} + u_{zz}$$

$$|\nabla u|^2 = |\text{grad} u|^2 = |\nabla u|^2 = u_x^2 + u_y^2 + u_z^2.$$

Let us recall again the divergence theorem:

$$\iiint_D \text{div} F dx dy dz = \iint_{\partial D} F \cdot n dS,$$

where F is any vector function, D is a bounded solid region, and n is the unit outer normal on ∂D.

Now we start from the product rule,

$$(v u_x)_x = v_x u_x + v u_{xx}$$

and the same with y and z derivatives. Summing, this leads to the identity

$$\nabla \cdot (v \nabla u) = \nabla v \cdot \nabla u + v \triangle u.$$

Then we integrate and use the divergence theorem on the left side to get the *Green's first identity*:

$$\boxed{\iint_{\partial D} v \frac{\partial u}{\partial n} dS = \iiint_D \nabla v \cdot \nabla u dx dy dz + \iiint_D v \triangle u dx dy dz,} \tag{4.2}$$

where $\frac{\partial u}{\partial n} = n \cdot \nabla u$ is the directional derivative in the outward normal direction.

4.2 Basic Properties of Harmonic Functions

4.2.1 Mean value property

Theorem 4.1 *The average value of any harmonic function u over any sphere equals its value at the center, i.e.,*

$$\boxed{u(M_0) = \frac{1}{4\pi r^2} \iint_{S_r(M_0)} u dS,} \tag{4.3}$$

where $S_r(M_0)$ is the sphere with radius r and center M_0.

Proof: Without loss of generality, we assume that $M_0 = O$. Let D be a ball with radius a, i.e., $D = \{x^2 + y^2 + z^2 < a^2\}$. Then $\partial D = \{x^2 + y^2 + z^2 = a^2\}$. Notice that along ∂D,

$$\frac{\partial u}{\partial n} = n \cdot \nabla u = \frac{x}{r} u_x + \frac{y}{r} u_y + \frac{z}{r} u_z = \frac{\partial u}{\partial r},$$

where $r = (x^2 + y^2 + z^2)^{1/2}$. By the Green's first identity (4.2) with $v \equiv 1$, we have

$$\iint_{\partial D} \frac{\partial u}{\partial r} dS = \iiint_D \triangle u dx dy dz = 0. \tag{4.4}$$

Let's write this integral in spherical coordinates (r, θ, w), where $x = r \sin\theta \cos w$, $y = r \sin\theta \sin w$, and $z = r\cos\theta$ for $0 \le w \le 2\pi$ and $0 \le \theta \le \pi$. Notice that $r = a$ on ∂D, and $dxdydz = drdS = r^2 \sin\theta dr d\theta dw$, so

$$\iint_{\partial D} \frac{\partial u}{\partial r} dS = \int_0^{2\pi} \int_0^{\pi} u_r(a, \theta, w) a^2 \sin\theta d\theta dw = 0,$$

for any $a > 0$. Hence

$$\frac{\partial}{\partial r} \left[\int_0^{2\pi} \int_0^{\pi} u(r, \theta, w) \sin\theta d\theta dw \right] = \int_0^{2\pi} \int_0^{\pi} u_r(a, \theta, w) \sin\theta d\theta dw = 0.$$

It means that the integral

$$\frac{1}{4\pi} \int_0^{2\pi} \int_0^{\pi} u(r, \theta, w) \sin\theta d\theta dw$$

is *independent* of r. Therefore

$$u(O) = \frac{1}{4\pi} \int_0^{2\pi} \int_0^{\pi} u(O) \sin\theta d\theta dw = \frac{1}{4\pi} \int_0^{2\pi} \int_0^{\pi} u(a, \theta, w) \sin\theta d\theta dw = \frac{1}{4\pi a^2} \iint_D u dS.$$

■

4.2.2 Maximum principle

Based on the mean value property, we can obtain the following powerful maximum principles.

Theorem 4.2 *If D is any solid region, i.e., D is a connected bounded open set with a smooth boundary ∂D, then a non-constant harmonic function in D cannot take its maximum value inside D, but only on ∂D.*

Proof: Assume the maximum value M of a harmonic function u is attained at $\mathbf{x}_M \in D$ (not on the boundary). By the definition of M, we know that

$$u(\mathbf{x}) \leq u(\mathbf{x}_M) \leq M, \qquad \text{for all } \mathbf{x} \in D.$$

We draw a surface around $\mathbf{x}_M$ entirely contained in D. By the mean value property,

$$M = u(\mathbf{x}_M) = \text{average on circle} \leq M.$$

Therefore, $u \equiv M$ for all $\mathbf{x}$ on the surface. So this is true for all $\mathbf{x}$ in a ball contained in D and with $\mathbf{x}_M$. Now we repeat the argument with a different center. We can fill the whole domain up with circles. By the assumption that D is connected, we deduce that $u \equiv M$ throughout D. ■

Theorem 4.3 (Hopf's Lemma) *If D is any solid region, then the outward normal derivative of a non-constant harmonic function u is strictly positive at a maximum point, i.e., $\partial u/\partial n > 0$.*

Proof: Assume that $u(x, y, z)$ attains its maximum value at the point $M_0 = (x_0, y_0, z_0) \in \partial D$. Also assume that there exists a ball $B_R \subset D$ with center (x_1, y_1, z_1) and radius R, such that $M_0 \in \partial B_R$, i.e., $(x_0-x_1)^2+(y_0-y_1)^2+(z_0-z_1)^2 = R^2$. Let $\Omega := \{(x, y, z) : R^2/4 \leq (x-x_1)^2+(y-y_1)^2+(z-z_1)^2 \leq R^2\}$. And in Ω, let

$$w(x, y, z) = u(x, y, z) - u(x_0, y_0, z_0) + \epsilon(e^{-a[(x-x_1)^2+(y-y_1)^2+(z-z_1)^2]} - e^{-aR^2}),$$

where $\epsilon > 0$ and $a > 0$ are two constants. By directly calculation,

$$\triangle w = w_{xx} + w_{yy} + w_{zz} = \epsilon\{-6a + 4a^2[(x-x_1)^2 + (y-y_1)^2 + (z-z_1)^2]\}e^{-a[(x-x_1)^2+(y-y_1)^2+(z-z_1)^2]}$$

Notice that in Ω, $R^2/4 \leq (x-x_1)^2+(y-y_1)^2+(z-z_1)^2 \leq R^2$, so we can choose $a > 0$ large enough such that $\triangle w > 0$ in Ω. If w attains its local maximum value inside Ω, then at this point $\triangle w = w_{xx} + w_{yy} + w_{zz} \leq 0$, which contradicts to the fact that $\triangle w > 0$ in Ω. Thus

$$\max_{\Omega} w = \max_{\partial\Omega} w.$$

On the boundary $(x-x_1)^2+(y-y_1)^2+(z-z_1)^2 = R^2/4$, Theorem 4.2 assures that $u(x, y, z)-u(x_0, y_0, z_0) < 0$. Hence we can choose $\epsilon > 0$ small enough such that $w < 0$ along it.

On the other boundary $(x-x_1)^2 + (y-y_1)^2 + (z-z_1)^2 = R^2$, obviously $w = u(x, y, z) - u(x_0, y_0, z_0) \leq 0$. Therefore $w \leq 0$ in Ω. Notice that at M_0, $w = 0$. So w attains its maximum value at M_0. Then at M_0,

$$\frac{\partial w}{\partial n} = \frac{\partial u}{\partial n} - 2\epsilon a R^2 e^{-aR^2} \geq 0.$$

It means

$$\frac{\partial u}{\partial n} \geq 2\epsilon a R^2 e^{-aR^2} > 0.$$

■

4.2.3 Dirichlet's principle

This is an important mathematical theorem based on the physical idea of energy. It states that among *all* the functions w in D that satisfy the Dirichlet boundary condition

$$\boxed{w = h \qquad \text{on } \partial D,} \tag{4.5}$$

the *lowest energy* occurs for the harmonic function satisfying (4.5). Here the energy is defined as

$$\boxed{E[w] := \frac{1}{2}\iiint_D |\nabla w|^2 dxdydz.} \tag{4.6}$$

This is the pure potential energy, there being no kinetic energy because there is no motion.

Theorem 4.4 *Let u be the harmonic function in D that satisfies* (4.5)*. Let w be any function in D that satisfies* (4.5)*. Then*

$$E[w] \geq E[u]. \tag{4.7}$$

Proof: First since $E[u-w] \geq 0$, we have

$$E[w] = \frac{1}{2}\iiint_D |\nabla[u-(u-w)]|^2 dxdydz \geq E[u] - \iiint_D \nabla u \cdot \nabla(u-w) dxdydz.$$

Now by the Green's first identity,

$$\iiint_D \nabla u \cdot \nabla(u-w) dxdydz = -\iiint_D (u-w)\triangle u dxdydz + \iint_{\partial D}(u-w)\frac{\partial u}{\partial n} dS.$$

Notice that $u - w = 0$ on ∂D, and $\triangle u = 0$ in D. So

$$E[w] \geq E[u] - \iiint_D \nabla u \cdot \nabla(u-w) dxdydz = E[u].$$

■

Exercises

1. Derive the two-dimensional Holf's Lemma by mimicking the proof of theorem 4.3.

2. Prove the stability of the Dirichlet problem with respect to the boundary conditions by using the maximum principle.

3. Prove the uniqueness up to constants of the Neumann problem by using the Holf's Lemma.

4. Prove Dirichlet's principle for the Neumann boundary condition. It asserts that among *all* real-valued functions $w(\mathbf{x})$ on D, the quantity

$$E[w] = \frac{1}{2}\iiint_D |\nabla w|^2 d\mathbf{x} - \iint_{\partial D} hw dS$$

is the smallest for $w = u$, where u is the solution of the Neumann problem

$$-\triangle u = 0 \quad \text{in } D, \qquad \frac{\partial u}{\partial n} = h(\mathbf{x}) \quad \text{on } \partial D.$$

Here the average of the given function $h(\mathbf{x})$ is zero.

4.3 Representation Formula

4.3.1 Green's second identity

Notice that the middle term in the Green's first identity (4.2) does not change if u and v are switched. So if we write (4.2) for the pair u and v, and again for the pair v and u, then subtract, we get the *Green's second*

identity

$$\boxed{\iiint_D (u\triangle v - v\triangle u)dxdydz = \iint_{\partial D}\left(u\frac{\partial v}{\partial n} - v\frac{\partial u}{\partial n}\right)dS.} \tag{4.8}$$

4.3.2 Fundamental solution

Before giving the representation formula, I would like to introduce a special solution which is called the *fundamental solution*.

It is natural to use *spherical coordinates* (r, θ, w), where $x = r\sin\theta\cos w$, $y = r\sin\theta\sin w$, and $z = r\cos\theta$ for $0 \le w \le 2\pi$ and $0 \le \theta \le \pi$. By the chain rule, obviously,

$$\triangle u = u_{xx} + u_{yy} + u_{zz} = u_{rr} + \frac{2}{r}u_r + \frac{1}{r^2}\left[u_{\theta\theta} + (\cot\theta)u_\theta + \frac{1}{\sin^2\theta}u_{ww}\right].$$

Finally, let's look for the special harmonic functions in three dimensions *which don't change under rotations*, that is, which depend only on r. Thus u satisfies the ODE

$$u_{rr} + \frac{2}{r}u_r = 0.$$

It is equal to $(r^2u)_r = 0$. It has the solutions $r^2u_r = C_1$, where C_1 is a constant. So $u = -C_1r^{-1} + C_2$. Now we get an important harmonic function, which is called the fundamental radial solution,

$$u(r) = \frac{1}{r} = \frac{1}{\sqrt{x^2+y^2+z^2}}.$$

It satisfies the Laplace equation everywhere except the origin, where it is *infinite*.

4.3.3 Representation formula

This formula represents any harmonic function as an integral over the boundary. It states as follows:

Theorem 4.5 *If $\triangle u = 0$ in D, then*

$$\boxed{u(\mathbf{x}_0) = \iint_{\partial D}\left[-u(\mathbf{x})\frac{\partial}{\partial n}\left(\frac{1}{|\mathbf{x}-\mathbf{x}_0|}\right) + \frac{1}{|\mathbf{x}-\mathbf{x}_0|}\frac{\partial u}{\partial n}\right]\frac{dS}{4\pi},} \tag{4.9}$$

where $\mathbf{x} = (x, y, z)$ and $\mathbf{x}_0 = (x_0, y_0, z_0)$.

Proof: Let

$$v(\mathbf{x}) = -\frac{1}{4\pi|\mathbf{x}-\mathbf{x}_0|}.$$

For simplicity, we let $\mathbf{x}_0$ be the origin O. Then $v = (-4\pi r)^{-1}$, where $r = |\mathbf{x}|$. Let D_ϵ be the region D with the ball (of radius ϵ and centre O) excised. Obviously, $\triangle u = \triangle v = 0$ in D_ϵ. So by the Green's second identity (4.8),

$$-\iint_{\partial D_\epsilon}\left[u\cdot\frac{\partial}{\partial n}\left(\frac{1}{r}\right) - \frac{\partial u}{\partial n}\cdot\frac{1}{r}\right]dS = 0.$$

Notice that the boundary ∂D_ϵ consists of two parts: the original boundary ∂D and the sphere $\{r = \epsilon\}$. On the sphere, $\partial/\partial n = -\partial/\partial r$. Therefore,

$$-\iint_{\partial D}\left[u\cdot\frac{\partial}{\partial n}\left(\frac{1}{r}\right)-\frac{\partial u}{\partial n}\cdot\frac{1}{r}\right]dS=-\iint_{r=\epsilon}\left[u\cdot\frac{\partial}{\partial r}\left(\frac{1}{r}\right)-\frac{\partial u}{\partial r}\cdot\frac{1}{r}\right]dS. \tag{4.10}$$

Now we are going to prove that the right hand side of (4.10) tends to $4\pi u(O)$ as $\epsilon \to 0$.

Notice that the right hand side of (4.10) equals

$$I_\epsilon := \frac{1}{\epsilon^2}\iint_{r=\epsilon} u dS + \frac{1}{\epsilon}\iint_{r=\epsilon}\frac{\partial u}{\partial r}dS. \tag{4.11}$$

Moreover,

$$\lim_\epsilon I_\epsilon = \lim_{\epsilon\to 0}\left[\frac{4\pi\epsilon^2 u(O)}{\epsilon^2}+\frac{1}{\epsilon^2}\iint_{r=\epsilon}(u-u(O))dS+\frac{4\pi\epsilon^2 u_r(O)}{\epsilon}+\frac{1}{\epsilon}\iint_{r=\epsilon}(u_r-u_r(O))dS\right]=4\pi u(O).$$

■

Remark 4.5.1 *The corresponding formula in* two dimensions *is*

$$\boxed{u(\mathbf{x}_0)=\iint_{\partial D}\left[u(\mathbf{x})\frac{\partial}{\partial n}\left(\log|\mathbf{x}-\mathbf{x}_0|\right)-\log|\mathbf{x}-\mathbf{x}_0|\frac{\partial u}{\partial n}\right]\frac{dS}{2\pi}.} \tag{4.12}$$

Notice that the two dimensional fundamental harmonic function is $u(r) = \log r$. Then the two-dimensional representation formula can be proved similarly.

4.4 Green Function

We now use Green's identities to study the Dirichlet problem.

Definition 4.6 *The* Green's function $G(\mathbf{x})$ *for the operator* $-\triangle$ *and the domain* D *at the point* $\mathbf{x}_0 \in D$ *is a function defined for* $\mathbf{x} \in D$ *such that:*

(i) $G(\mathbf{x})$ *possesses continuous second derivatives and* $\triangle G = 0$ *in* D*, except at the point* $\mathbf{x} = \mathbf{x}_0$*.*

(ii) $G(\mathbf{x}) = 0$ *for* $\mathbf{x} \in \partial D$*.*

(iii) The function $G(\mathbf{x}) + \frac{1}{4\pi|\mathbf{x}-\mathbf{x}_0|}$ *is finite at* $\mathbf{x}_0$ *and has continuous second derivatives everywhere and is harmonic at* $\mathbf{x}_0$*.*

Remark 4.6.1 *In two dimension, we only need to change the condition* (*iii*) *to the condition that* the function $G(\mathbf{x}) - \frac{1}{2\pi}\log|\mathbf{x}-\mathbf{x}_0|$ is finite at $\mathbf{x}_0$ and has continuous second derivatives everywhere and is harmonic at $\mathbf{x}_0$.

It can be shown that a Green's function exists. Also it is unique. The usual notation for the Green's function is $G(\mathbf{x}, \mathbf{x}_0)$.

Theorem 4.7 *If $G(\mathbf{x}, \mathbf{x}_0)$ is the Green's function, then the solution of the Dirichlet problem is given by the formula*

$$\boxed{u(\mathbf{x}_0) = \iint_{\partial D} u(\mathbf{x}) \frac{\partial G(\mathbf{x}, \mathbf{x}_0)}{\partial n} dS.} \tag{4.13}$$

Proof: First from the representation formula (4.9),

$$u(\mathbf{x}_0) = \iint_{\partial D} \left[u(\mathbf{x}) \frac{\partial v}{\partial n} - \frac{\partial u}{\partial n} v \right] dS,$$

where $v(\mathbf{x}) = -\frac{1}{4\pi|\mathbf{x}-\mathbf{x}_0|}$. Let

$$H(\mathbf{x}) := G(\mathbf{x}, \mathbf{x}_0) - v(\mathbf{x}).$$

Then from definition 4.6, $H(\mathbf{x})$ is a harmonic function throughout the domain D. Now by the Green's second identity,

$$0 = \iint_{\partial D} \left[u(\mathbf{x}) \frac{\partial H}{\partial n} - \frac{\partial u}{\partial n} H \right] dS.$$

Adding together, we get

$$u(\mathbf{x}_0) = \iint_{\partial D} \left[u(\mathbf{x}) \frac{\partial v}{\partial n} - \frac{\partial u}{\partial n} v \right] dS + \iint_{\partial D} \left[u(\mathbf{x}) \frac{\partial H}{\partial n} - \frac{\partial u}{\partial n} H \right] dS = \iint_{\partial D} \left[u(\mathbf{x}) \frac{\partial G}{\partial n} - \frac{\partial u}{\partial n} G \right] dS.$$

Notice that G vanishes on ∂D, so the last term vanishes and we complete the proof. ■

The Green's function also allows us to solve *Poisson's equation*.

Theorem 4.8 *The solution of the problem*

$$\triangle u = f \quad \text{in } D, \qquad u = h \quad \text{on } \partial D \tag{4.14}$$

is given by

$$\boxed{u(\mathbf{x}_0) = \iint_{\partial D} h(\mathbf{x}) \frac{\partial G(\mathbf{x}, \mathbf{x}_0)}{\partial n} dS + \iiint_D f(\mathbf{x}) G(\mathbf{x}, \mathbf{x}_0) d\mathbf{x}.} \tag{4.15}$$

Remark 4.8.1 *The Poisson formula* (4.15) *is also valid for two dimensional domains D.*

Proof: Notice that now $\triangle u = f$ in D. Then following the same procedure in the proof of theorem 4.5, we have the following representation theorem

$$u(\mathbf{x}_0) = \iint_{\partial D} \left[-u(\mathbf{x}) \frac{\partial}{\partial n} \left(\frac{1}{|\mathbf{x} - \mathbf{x}_0|} \right) - \frac{1}{|\mathbf{x} - \mathbf{x}_0|} \frac{\partial u}{\partial n} \right] \frac{dS}{4\pi} - \iiint_D f(\mathbf{x}) \frac{1}{4\pi|\mathbf{x} - \mathbf{x}_0|} d\mathbf{x}, \tag{4.16}$$

Let $H(\mathbf{x}) := G(\mathbf{x}, \mathbf{x}_0) - v(\mathbf{x})$. Again from definition 4.6, $H(\mathbf{x})$ is a harmonic function throughout the domain D. Now by the Green's second identity,

$$0 = \iint_{\partial D} \left[u(\mathbf{x}) \frac{\partial H}{\partial n} - \frac{\partial u}{\partial n} H(\mathbf{x}) \right] dS + \iiint_D f(\mathbf{x}) H(\mathbf{x}) d\mathbf{x}.$$

Hence summing together, we can get (4.15) ■

4.4.1 Green functions in the half-space

We first determine the Green's function for a half-space with the *method of reflection*. A half-space is the region lying on one side of a plane. We write the coordinates as $\mathbf{x} = (x, y, z)$. Say that the half-space is $D = \{z > 0\}$, the domain that lies above the xy plane. Notice that each point $\mathbf{x} = (x, y, z)$ in D has a *reflected point* $\mathbf{x}^* = (x, y, -z)$ that is not in D. From the fundamental solution, we assert that the Green's function for D is

$$\boxed{G(\mathbf{x}, \mathbf{x}_0) = -\frac{1}{4\pi|\mathbf{x} - \mathbf{x}_0|} + \frac{1}{4\pi|\mathbf{x} - \mathbf{x^*}_0|}.} \tag{4.17}$$

Let's verify the assertion by checking each of the three properties of G in definition 4.6.

(i) Clearly, G is finite and differentiable except at $\mathbf{x}_0$. Also, $\triangle G = 0$.

(ii) Because $\mathbf{x^*}_0$ is outside the half-space D, the function $\frac{1}{4\pi|\mathbf{x}-\mathbf{x^*}_0|}$ has no singularity inside the domain, so that G has the proper singularity at $\mathbf{x}_0$.

(iii) Let $\mathbf{x} \in \partial D$, so $z = 0$. Hence $|\mathbf{x} - \mathbf{x}_0| = |\mathbf{x} - \mathbf{x^*}_0| = \sqrt{(x - x_0)^2 + (y - y_0)^2 + z_0^2}$. Thus $G(\mathbf{x}, \mathbf{x}_0) = 0$.

Remark 4.8.2 *Similarly for two dimensional case, the Green's function for the half-plane $D = \{y > 0\}$ is*

$$\boxed{G(\mathbf{x}, \mathbf{x}_0) = -\frac{1}{2\pi}\log|\mathbf{x} - \mathbf{x}_0| + \frac{1}{2\pi}\log|\mathbf{x} - \mathbf{x^*}_0|.} \tag{4.18}$$

Based on the Green's function in the half-space, we can solve the Dirichlet problem

$$\boxed{\triangle u = 0 \quad \text{for } z > 0, \qquad u(x, y, 0) = h(x, y).} \tag{4.19}$$

Notice that along ∂D, $\frac{\partial G}{\partial n} = -\frac{\partial G}{\partial z}\big|_{z=0}$. And

$$-\frac{\partial G}{\partial z} = \frac{1}{4\pi}\left(\frac{z + z_0}{|\mathbf{x} - \mathbf{x^*}_0|^3} - \frac{z - z_0}{|\mathbf{x} - \mathbf{x^*}_0|^3}\right) = \frac{1}{2\pi}\frac{z_0}{|\mathbf{x} - \mathbf{x^*}_0|^3}.$$

Therefore, we have

$$\boxed{u(\mathbf{x}_0) = \frac{z_0}{2\pi}\iint_{\partial D}\frac{h(\mathbf{x})}{|\mathbf{x} - \mathbf{x^*}_0|^3}dS.} \tag{4.20}$$

4.4.2 Green functions in the ball

The Green's function for the ball $D = \{|\mathbf{x}| < a\}$ of radius a can also be found by the *reflection method*. Fix any point $\mathbf{x}_0$ in the ball (that is, $|\mathbf{x}_0| < a$). The *reflected point* $\mathbf{x^*}_0$ is defined such that it is collinear with the origin O and the point $\mathbf{x}_0$, while its distance from the origin is determined by the formula $|\mathbf{x}_0||\mathbf{x^*}_0| = a^2$. Therefore

$$\mathbf{x^*}_0 = \frac{a^2\mathbf{x}_0}{|\mathbf{x}_0|^2}. \tag{4.21}$$

Hence the Green's function of the ball is

$$\boxed{G(\mathbf{x}, \mathbf{x}_0) = -\frac{1}{4\pi|\mathbf{x} - \mathbf{x}_0|} + \frac{a}{|\mathbf{x}_0|}\frac{1}{4\pi|\mathbf{x}_0 - \mathbf{x^*}_0|}.} \tag{4.22}$$

Let's verify the assertion by checking each of the three properties of G in definition 4.6.

First of all, G has n singularity except at $\mathbf{x} = \mathbf{x}_0$, because $\mathbf{x}^*{}_0$ lies outside of the ball. Hence (i) and (iii) are obviously true.

To prove (ii), for any $\mathbf{x} \in \partial D$ (that is, $|\mathbf{x}| = a$), the triangle $O\mathbf{x}\mathbf{x}_0$ is congruent to the triangle $O(\frac{|\mathbf{x}_0|}{a}\mathbf{x})(\frac{a}{|\mathbf{x}_0|}\mathbf{x}_0)$. It is because that $\frac{a}{|\mathbf{x}_0|}\mathbf{x}_0 \in \partial D$. Therefore

$$\left|\frac{|\mathbf{x}_0|}{a}\mathbf{x} - \frac{a}{|\mathbf{x}_0|}\mathbf{x}_0\right| = |\mathbf{x} - \mathbf{x}_0|. \tag{4.23}$$

Thus

$$|\mathbf{x} - \mathbf{x}_0| = \frac{|\mathbf{x}_0|}{a}\left|\mathbf{x} - \frac{a^2}{|\mathbf{x}_0|^2}\mathbf{x}_0\right| = \frac{|\mathbf{x}_0|}{a}|\mathbf{x} - \mathbf{x}^*{}_0|. \tag{4.24}$$

It means that $G(\mathbf{x}, \mathbf{x}_0) = 0$ for any $\mathbf{x} \in \partial D$.

Based on the Green's function for the ball, we can solve the following Dirichlet problem

$$\boxed{\triangle u = 0 \quad \text{in } |\mathbf{x}| < a, \qquad u = h \quad \text{on } |\mathbf{x}| = a.} \tag{4.25}$$

Obviously, we only need to calculate $\frac{\partial G}{\partial n}$ on $|\mathbf{x}| = a$. First

$$\nabla G = \frac{\mathbf{x} - \mathbf{x}_0}{4\pi|\mathbf{x} - \mathbf{x}_0|^3} - \frac{a}{|\mathbf{x}_0|}\frac{\mathbf{x} - \mathbf{x}^*{}_0}{4\pi|\mathbf{x} - \mathbf{x}^*{}_0|^3}.$$

Notice that if $|\mathbf{x}| = a$, then we have that $|\mathbf{x} - \mathbf{x}^*{}_0| = \frac{a}{|\mathbf{x}_0|}|\mathbf{x} - \mathbf{x}_0|$. Now by $\mathbf{x}^*{}_0 = \frac{a^2}{|\mathbf{x}_0|^2}\mathbf{x}_0$,

$$\nabla G = \frac{1}{4\pi|\mathbf{x} - \mathbf{x}_0|^3}\left[\mathbf{x} - \mathbf{x}_0 - \frac{|\mathbf{x}_0|^2}{a^2}\mathbf{x} + \mathbf{x}_0\right]$$

when $|\mathbf{x}| = a$. So along ∂D,

$$\frac{\partial G}{\partial n} = \frac{\mathbf{x}}{a} \cdot \nabla G = \frac{a^2 - |\mathbf{x}_0|^2}{4\pi a|\mathbf{x} - \mathbf{x}_0|^3}.$$

Thus the solution to the Dirichlet problem (4.25) is

$$\boxed{u(\mathbf{x}_0) = \frac{a^2 - |\mathbf{x}_0|^2}{4\pi a}\iint_{|\mathbf{x}|=a}\frac{h(\mathbf{x})}{|\mathbf{x} - \mathbf{x}_0|^3}dS.} \tag{4.26}$$

We remark here that if $\mathbf{x}_0 = O$, then $u(O) = \frac{1}{4\pi a^2}\iint_{|\mathbf{x}|=a} h(\mathbf{x})dS$. It is exactly the mean value property.

Exercises

1. Let $\phi(\mathbf{x})$ be any smooth function defined on all of three-dimensional space that vanishes outside some sphere. Show that

$$\phi(O) = -\iiint \frac{1}{|\mathbf{x}|}\triangle\phi(\mathbf{x})\frac{d\mathbf{x}}{4\pi}.$$

The integration is taken over the region where $\phi(\mathbf{x})$ is not zero.

2. Show that the Green's function is unique. (Hint: Take the difference of two of them.)

3. Show that the Green's function is always symmetric: $G(\mathbf{x}, \mathbf{x}_0) = G(\mathbf{x}_0, \mathbf{x})$ for $\mathbf{x} \neq \mathbf{x}_0$.

4. Find the one-dimensional Green's function for the interval $(0, l)$. The three properties defining is

(i) It solves $G''(x) = 0$ for $x =\neq x_0$.

(ii) $G(0) = G(l) = 0$.

(iii) G(x) is continuous at x_0 and $G(x) + \frac{1}{2}|x - x_0|$ is harmonic at x_0.

5. (a) Find the Green's function for the half-plane $\{(x, y) : y > 0\}$.

(b) Use it to solve the Dirichlet problem in the half-plane with boundary values $h(x)$.

(c) Calculate the solution with $u(x, 0) = 1$.